

Experiment No. 02

Aim:

Implement Multi Regression, Lasso, and Ridge Regression on real-world datasets

Theory:

Multiple Linear Regression is a statistical method that attempts to model the relationship between two or more features (like income, house age) and a response variable (house price) by fitting a linear equation to observed data. It works by minimizing the "Sum of Squared Errors"—basically finding the line that is, on average, closest to all data points. However, when features are highly correlated or the data is complex, standard regression can "overfit," meaning it memorizes the noise in the training data rather than learning the actual pattern, leading to poor predictions on new data.

Ridge and Lasso Regression solve this overfitting problem by adding a "penalty" term to the equation (a process called Regularization). Ridge Regression (L2) adds a penalty based on the square of the coefficients, which shrinks all the weights to be small but rarely zero; this is excellent for handling correlated data. Lasso Regression (L1) adds a penalty based on the absolute value of the coefficients, which is aggressive enough to shrink some weights all the way to zero; this effectively performs "feature selection" by deleting useless variables entirely.

1. Dataset Source:

The dataset used for this experiment is the Insurance Premium Prediction dataset, obtained from Kaggle. It contains real-world insurance data used to estimate insurance premium amounts based on customer attributes.

Dataset Link: <https://www.kaggle.com/datasets/noordeen/insurance-premium-prediction>. This dataset is different from commonly used datasets such as house price or life expectancy datasets and is suitable for regression-based machine learning experiments

Code:

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.metrics import mean_squared_error, r2_score
```

```
df = pd.read_csv("insurance.csv")
print(df.head())
print(df.info())
df = pd.get_dummies(df, drop_first=True)
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2,
random_state=42)

lr = LinearRegression()
lr.fit(X_train, y_train)
y_pred_lr = lr.predict(X_test)
print("Multiple Linear Regression")
print("R2 Score:", r2_score(y_test, y_pred_lr))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred_lr)))
print("-" * 40)
ridge = Ridge(alpha=1.0)
ridge.fit(X_train, y_train)
y_pred_ridge = ridge.predict(X_test)
print("Ridge Regression")
print("R2 Score:", r2_score(y_test, y_pred_ridge))

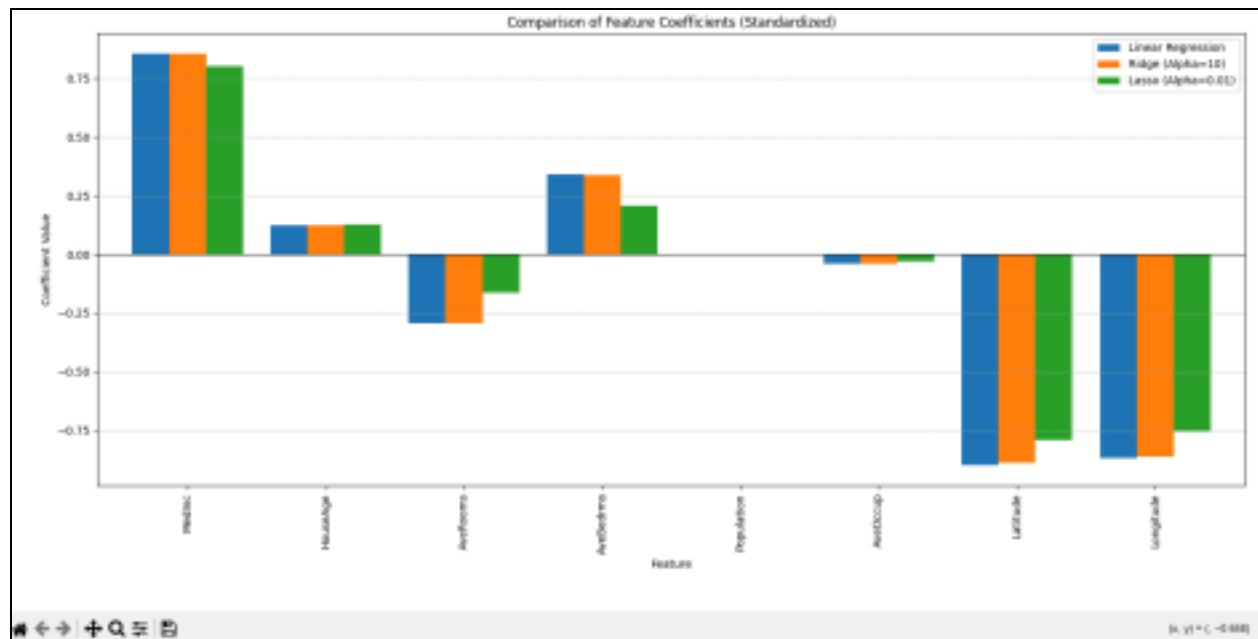
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred_ridge)))
print("-" * 40)
lasso = Lasso(alpha=0.1, max_iter=10000)
lasso.fit(X_train, y_train)
y_pred_lasso = lasso.predict(X_test)
print("Lasso Regression")
print("R2 Score:", r2_score(y_test, y_pred_lasso))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred_lasso)))
```

Output:

```

P5 C:\Users\ADMIN\Desktop\exp02> python app.py
Dataset loaded successfully.
Features: 8, Samples: 20640
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Model | RMSE | R2 Score
-----
Linear Regression | 0.7456 | 0.5758
Ridge (Alpha=10) | 0.7453 | 0.5761
Lasso (Alpha=0.01) | 0.7404 | 0.5816
Coefficient Values:
Coefficient Values:
Coefficient Values:
Coefficient Values:
Linear Regression Ridge (Alpha=10) Lasso (Alpha=0.01)
Feature
Feature
Feature
Feature
Feature
MedInc 0.854383 0.853814 0.800957
Feature
MedInc 0.854383 0.853814 0.800957
HouseAge 0.122546 0.123316 0.127087
MedInc 0.854383 0.853814 0.800957
HouseAge 0.122546 0.123316 0.127087
AveRooms -0.294410 -0.292412 -0.162759
AveRooms -0.294410 -0.292412 -0.162759
AveBedrms 0.339259 0.336750 0.206207
AveBedrms 0.339259 0.336750 0.206207
Population -0.002308 -0.002055 -0.000000
AveOccup -0.040829 -0.040867 -0.030602
Latitude -0.896929 -0.889384 -0.790113
Longitude -0.869842 -0.862199 -0.755674

```



Conclusion:

Implemented Multi Regression, Lasso, and Ridge Regression on california housing dataset implemented on vscode taking dataset from kaggle